Teacher Retention and End-of-Course Examination Outcomes: Evidence from Eight Greenville County, South Carolina, High Schools, 2022–23 to 2023–24

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Abstract. **This study asks how teacher-return rates and End-of-Course Examination Program (EOCEP) outcomes changed at eight Greenville County, South Carolina, high schools from 2022–23 to 2023–24. It uses school-level English II and Algebra I results from the South Carolina School Report Cards, measured as the percentage of students earning a C or higher, and compares each school's change with the district and state. Title I status is treated as context rather than a direct measure of poverty. Results varied across schools and subjects. Berea and Southside improved relative to district change in both subjects. Carolina gained 9.8 points in English II but declined 0.3 points in Algebra I. Wade Hampton's English II rate fell 1.9 points, while its Algebra I rate rose 0.8 points, 8.6 points behind the district's 9.4-point gain. Teacher-return rates also varied, and the eight-school pattern does not support a simple one-to-one explanation of exam outcomes by retention. Across the eight schools, 2023–24 poverty percentages correlated with pass rates at r = −0.9647 for English II and r = −0.9556 for Algebra I, but because the schools fall into two distinct socioeconomic groups, these figures largely reflect the gap between those groups rather than a stable relationship across schools. The findings are descriptive, not causal. The clearest follow-up priority is a focused, supportive review of Wade Hampton's Algebra I result, and district leaders should track retention alongside subject-level results rather than use retention as a stand-alone indicator.**

Keywords. teacher retention; EOCEP; student achievement; Title I; Greenville County

1. Introduction

Teacher turnover can disrupt classroom routines, professional relationships, and school-improvement work. Research has associated teacher turnover with adverse student-achievement effects in some settings, while also showing that consequences depend on which teachers leave, how vacancies are filled, and the conditions in which schools operate. Retention is therefore a useful organizational indicator, but it should not be treated as a stand-alone explanation for test results.

District-wide averages can conceal differences among schools and subjects. A school may improve in one EOCEP subject while declining in another, and a higher teacher-return rate may coincide with either stronger or weaker outcomes. This study places teacher-return information alongside English II and Algebra I results to describe those patterns.

The analysis focuses on Berea, Carolina, Eastside, Riverside, Wade Hampton, J.L. Mann, Mauldin, and Southside. It compares 2022–23 with 2023–24 and does not incorporate later results into the core change calculations. The purpose is to identify evidence-based follow-up priorities, not to claim that teacher retention caused achievement gains or declines.

The research question is: how did English II and Algebra I EOCEP outcomes and teacher-return rates change at these eight Greenville County high schools between 2022–23 and 2023–24, and what patterns can inform school-level staffing and instructional support?

2. Literature Review

Ronfeldt, Loeb, and Wyckoff (2013) found that teacher turnover was associated with lower student achievement in their study of New York City schools, including patterns extending beyond the classrooms of departing teachers. Their findings provide a rationale for examining continuity, but they cannot automatically be transferred to every school or district.

Sorensen and Ladd (2020) examined costs of teacher turnover that extend beyond recruitment and replacement, including disruptions to school organization and the loss of accumulated professional knowledge. Adnot et al. (2017), using evidence from Washington, DC public schools, show why it matters who leaves and who replaces them: turnover does not have one uniform effect

in every staffing context. Tan, Wei, Carver-Thomas, and García (2026) place teacher movement and departure in a broader account of workforce conditions and turnover patterns.

These studies use broader or more detailed evidence than the present eight-school, two-year comparison. They support examining retention as one relevant factor, not treating the Greenville County results as proof of a causal relationship. Working-condition surveys may help explain why teachers stay or leave, but perceptions of working conditions are not themselves teacher-retention rates.

The conceptual expectation is not that higher retention must always produce higher EOCEP pass rates in the same year. Achievement also reflects student readiness, course enrollment, instructional supports, leadership, curriculum, cohort composition, and other school conditions. This analysis identifies patterns for further investigation rather than testing a causal model.

3. Data and Methods

The study uses school-level EOCEP results for 2022–23 and 2023–24 published through the South Carolina Department of Education (SCDE) School Report Cards, and teacher-return figures assembled in the project's teacher-return workbook. District and state values are included as benchmarks. The eight schools were identified by SCDE school IDs: Berea (2301002), Carolina (2301005), Eastside (2301006), Riverside (2301017), Wade Hampton (2301010), J.L. Mann (2301013), Mauldin (2301014), and Southside (2301018).

Outcome measure. The EOCEP pass rate is the percentage of students in the All Students category earning grades A, B, or C in English II or Algebra I, taken as the sum of the reported A, B, and C percentages. Values are reported to one decimal place.

Calculations. Percentage-point change equals the 2023–24 rate minus the 2022–23 rate. District-relative change equals the school's percentage-point change minus the district's change. Both are the authors' own calculations, not SCDE figures. A positive district-relative value means that the school's year-over-year change exceeded the district's change; it does not mean that the school's absolute pass rate was higher than the district's.

Teacher-return rate. Teacher-return rates are reproduced from the project-compiled workbook and compared descriptively with achievement changes. The workbook records the rates but not the counts behind them, so the teacher-return measure could not be independently recalculated from personnel records. This is treated as a data limitation.

Title I status. Title I status identifies schools served through a federal program intended to support students from low-income backgrounds. It is used as context only, not as a direct measure of economic disadvantage or a precise poverty ranking. Classification follows the SCDE year-specific list.

Poverty measure. The poverty measure is "Percent of students in poverty," reported in the South Carolina School Report Cards under Environment → Financial Data for the 2023–24 school year. The eight values and their report-card links are listed in Appendix A. The Pearson correlations between these percentages and the 2023–24 EOCEP C-or-higher rates are the authors' own calculation.

Interpretation. The analysis is descriptive. It does not estimate a causal effect of teacher retention, and the school-level results should not be interpreted as individual student or teacher outcomes.

4. Results

Tables I and II report the percentage of students earning a C or higher. Changes are expressed in percentage points (pp), not percent change.

4.1 English II

The district English II C-or-higher rate rose from 71.0% in 2022–23 to 71.5% in 2023–24, a 0.5-point increase. The state rate rose from 64.4% to 66.4%, a 2.0-point increase. School-level changes ranged from a 9.8-point gain at Carolina to a 2.1-point decline at J.L. Mann. Carolina, Southside, and Berea improved more than the district, with district-relative gains of 9.3, 6.6, and 4.1 points, respectively. Mauldin's increase was close to the district change. Eastside was almost unchanged, while Riverside, Wade Hampton, and J.L. Mann declined. Wade Hampton's English II rate fell 1.9 points, 2.4 points below the district's year-over-year change.

TABLE I. ENGLISH II EOCEP C-OR-HIGHER RATES, CHANGES IN PERCENTAGE POINTS

School	2022–23	2023–24	Change	vs. district
Berea	50.8%	55.4%	+4.6	+4.1
Carolina	49.4%	59.2%	+9.8	+9.3
Eastside	78.7%	79.0%	+0.3	−0.2
Riverside	84.4%	83.9%	−0.5	−1.0
Wade Hampton	76.5%	74.6%	−1.9	−2.4
J.L. Mann	74.6%	72.5%	−2.1	−2.6
Mauldin	79.3%	80.2%	+0.9	+0.4
Southside	48.8%	55.9%	+7.1	+6.6
Greenville County	**71.0%**	**71.5%**	**+0.5**	**n/a**
South Carolina	**64.4%**	**66.4%**	**+2.0**	**n/a**

Source: SCDE School Report Cards, EOCEP results, 2022–23 and 2023–24. Change columns are the authors' calculations.

4.2 Algebra I

The district Algebra I C-or-higher rate increased from 51.5% to 60.9%, a 9.4-point gain. The state rate increased from 41.8% to 47.8%, a 6.0-point gain. Seven of the eight schools improved; Carolina was the only school with a decline. Berea had the largest gain (+16.8 points), followed by Eastside (+13.5), J.L. Mann (+12.2), and Southside (+11.6). Each exceeded the district's 9.4-point increase. Riverside improved by 8.0 points, 1.4 points below the district change. Mauldin gained 4.1 points. Wade Hampton gained 0.8 points, 8.6 points below the district increase. Carolina had the largest relative shortfall, at 9.7 points below district change.

TABLE II. ALGEBRA I EOCEP C-OR-HIGHER RATES, CHANGES IN PERCENTAGE POINTS

School	2022–23	2023–24	Change	vs. district
Berea	27.7%	44.5%	+16.8	+7.4
Carolina	31.5%	31.2%	−0.3	−9.7
Eastside	52.0%	65.5%	+13.5	+4.1
Riverside	65.2%	73.2%	+8.0	−1.4
Wade Hampton	69.1%	69.9%	+0.8	−8.6
J.L. Mann	50.4%	62.6%	+12.2	+2.8
Mauldin	68.9%	73.0%	+4.1	−5.3
Southside	31.1%	42.7%	+11.6	+2.2
Greenville County	**51.5%**	**60.9%**	**+9.4**	**n/a**
South Carolina	**41.8%**	**47.8%**	**+6.0**	**n/a**

Source: SCDE School Report Cards, EOCEP results, 2022–23 and 2023–24. Change columns are the authors' calculations.

4.3 Cross-subject patterns

Carolina's strong English II gain occurred alongside a slight Algebra I decline. Eastside and J.L. Mann outpaced the district in Algebra I but lagged it in English II. Mauldin nearly matched the district's English II change but fell well short of its Algebra I gain. Berea and Southside improved relative to the district in both subjects. Wade Hampton's English II rate declined slightly, while its Algebra I rate increased modestly; the Algebra I gain nevertheless lagged substantially behind the district increase.

4.4 Teacher-return rates

Teacher-return rates varied across schools (Table III). Berea (+19.2 points), Carolina (+9.3), Riverside (+4.8), and Wade Hampton (+4.1) recorded increases; Eastside (−8.6), Mauldin (−6.0), J.L. Mann (−1.6), and Southside (−1.0) recorded decreases.

The two indicators did not move together consistently. Berea combined a large increase in teacher return with gains in both subjects. Carolina also increased teacher return and made a substantial English II gain, but Algebra I was slightly lower. Eastside's return rate declined while Algebra I improved strongly. Mauldin's return rate declined, yet both subject rates increased. Wade Hampton's reported teacher-return rate rose while English II declined slightly and Algebra I increased modestly. Southside and J.L. Mann also show mixed combinations of retention change and subject outcomes.

TABLE III. TEACHER-RETURN RATES AND EOCEP CHANGE, IN PERCENTAGE POINTS

School	Return 2022–23	Return 2023–24	Change	EOCEP change
Berea	69.2%	88.4%	+19.2	English II +4.6; Algebra I +16.8
Carolina	73.1%	82.4%	+9.3	English II +9.8; Algebra I −0.3
Eastside	92.1%	83.5%	−8.6	English II +0.3; Algebra I +13.5
Riverside	88.1%	92.9%	+4.8	English II −0.5; Algebra I +8.0
Wade Hampton	87.8%	91.9%	+4.1	English II −1.9; Algebra I +0.8
J.L. Mann	88.5%	86.9%	−1.6	English II −2.1; Algebra I +12.2
Mauldin	92.0%	86.0%	−6.0	English II +0.9; Algebra I +4.1
Southside	84.3%	83.3%	−1.0	English II +7.1; Algebra I +11.6

Source: teacher-return rates from the project teacher-return workbook; EOCEP changes from Tables I and II.

4.5 Title I designation

The project's year-specific classification identified Berea and Carolina as Title I schools. Southside was not designated Title I in 2022–23 and was designated Title I in 2023–24. The other five schools were classified as non-Title I in the records reviewed for the project. Because Southside's designation changed, a single Title I/non-Title I label cannot accurately represent both years.

4.6 Poverty and pass rates

The 2023–24 poverty percentages fall into two groups (Appendix A). Carolina (86.9%), Berea (81.5%), and Southside (78.7%) were above 78%; the other five schools were between 31.4% and 52.0%. Across the eight schools, the Pearson correlation between poverty and the 2023–24 C-or-higher rate was $r = -0.9647$ for English II and $r = -0.9556$ for Algebra I.

5. DISCUSSION

The clearest priority for follow-up is Wade Hampton's Algebra I result relative to the district. Its Algebra I rate rose by 0.8 percentage points, while the district increased by 9.4 points, leaving Wade Hampton 8.6 points behind the district's year-over-year change. Its English II rate declined by 1.9 points while the district rose 0.5 points, a relative shortfall of 2.4 points. The reported teacher-return rate increased over the same period, so the pattern cannot be explained simply by a decline in the number of returning teachers. A useful review would examine course-level staffing and vacancies, the experience and assignments of incoming and returning teachers, curriculum implementation, student attendance and preparation, and the timing and availability of instructional support. These are questions for investigation, not assumed causes.

A second finding is that the district's Algebra I increase was much larger than its English II increase. As a result, schools could post a positive Algebra I change and still fall behind the district. Mauldin's Algebra I rate rose 4.1 points, 5.3 points below the district's increase. Relative change is a useful complement to a school's own percentage-point change, but it should always be read alongside starting and ending rates.

The mixed retention patterns in Section 4.4 show why retention alone should not be used to explain short-term achievement shifts. Teacher return may matter through continuity and accumulated expertise, but school-level rates do not tell us which teachers returned, which courses they taught, the experience or effectiveness of replacement staff, or whether changes in tested student cohorts affected outcomes.

Berea and Carolina had relatively low starting EOCEP rates in some subjects and low teacher-return rates in 2022–23 compared with most schools in the sample. Both later recorded higher teacher-return rates. These overlapping patterns do not establish that retention caused subsequent achievement changes. Schools with low starting outcomes may have more room to improve, so baseline differences should be considered in future analysis rather than treated as an established explanation.

The results do not show that Title I status caused any outcome. Berea and Southside improved relative to the district in both subjects, while Carolina's improvement was concentrated in English II. The poverty correlations look strong, but because the eight schools fall into two distinct socioeconomic groups, they largely reflect the gap between those groups rather than a stable continuous relationship across schools. They should not be read as evidence that each incremental increase in poverty predicts a fixed decrease in pass rates, or as evidence of causation.

Retention and achievement are related policy concerns but not interchangeable measures. The results argue for a school-specific response: combine retention monitoring with subject-level achievement review, course and grade data, staffing assignments, and working-condition evidence. District leaders should avoid using a single retention target as the sole indicator of school improvement.

6. Recommendations

These recommendations are for Greenville County Schools district leaders and the leadership teams at the eight schools.

- *Conduct a focused, supportive review at Wade Hampton.* Review Algebra I performance relative to the district alongside English II results, course-level staffing, vacancies, teacher experience and assignments, attendance, student readiness, curriculum implementation, and available instructional support. The purpose should be diagnostic, not punitive.
- *Track retention and achievement in a shared school dashboard.* Show annual teacher-return rates beside subject-specific EOCEP rates and district-relative changes. Preserve the year, denominator, definition, and source for each metric.
- *Investigate subject-specific needs rather than applying one school-wide explanation.* Carolina's English II gain and Algebra I decline, along with Eastside and J.L. Mann's differing subject patterns, suggest that supports should align with course-level evidence.
- *Strengthen retention supports where rates decline.* For Eastside and Mauldin, examine exit reasons, vacancy duration, workload, mentoring, leadership support, and course-assignment patterns. Consider practical, locally designed incentives only where evidence supports them, and evaluate interventions rather than assuming they work.
- *Improve economic-context measures.* Pair Title I designation with verified school-level poverty or meal-eligibility data where possible. Document Southside's designation change rather than applying one label to both years.
- *Extend the analysis over more years.* Add additional cohorts and, where feasible, examine grade- and course-level results alongside teacher assignments. Longer time series can help distinguish temporary fluctuations from persistent patterns.
- *Preserve transparent evidence checks.* Keep source links, calculation notes, data dictionaries, and records of unavailable or suppressed values with project files.
- *Interpret the poverty-achievement correlations cautiously.* The coefficients are based on eight school-level observations and largely reflect the gap between two poverty groups. Future analyses should examine patterns within comparable poverty groups before drawing broader conclusions.

7. Limitations

- *No causal claims.* This study is descriptive. It cannot establish causation or determine whether individual students were taught by teachers who stayed or left. Results may be affected by changes in tested cohorts, student attendance and preparation, course enrollment, curriculum, leadership, teacher experience, hiring, vacancies, and other school conditions.
- *Sample size.* The study covers eight schools over two school years, using school-level aggregates only.
- *Teacher-return data.* Teacher-return percentages depend on the definition and denominator used in the underlying source. The project workbook contains rates for all eight schools, but a complete raw-data reconstruction of the teacher-return measure was not available. A returning-teacher measure does not capture teacher effectiveness, the experience of staff who leave or arrive, or differences between subject areas.
- *Title I as a proxy.* Title I designation is an imperfect proxy for economic disadvantage, and Southside's designation changed between the two years.
- *Poverty correlations.* The correlations are reproducible from the eight poverty values in Appendix A and the 2023–24 pass rates, but they rest on a small, socioeconomically clustered sample.

8. Conclusion

Across eight Greenville County high schools, teacher-return rates and EOCEP outcomes changed in different ways from 2022–23 to 2023–24. Berea and Southside gained ground relative to district change in both English II and Algebra I. Carolina's strongest

gain was in English II, while Eastside and J.L. Mann outpaced the district in Algebra I. Mauldin's English II change was close to the district's, while its Algebra I increase lagged behind. Wade Hampton's English II rate declined slightly and its Algebra I rate rose modestly, but that Algebra I gain was substantially smaller than the district increase.

The central policy implication is to use retention as one part of a wider school-improvement picture. Wade Hampton merits a focused, supportive review of its Algebra I result relative to district change, while other schools need responses tailored to their subject-specific patterns. A stronger follow-up study should connect verified teacher-return data to course-level staffing and achievement, include more years, and measure economic disadvantage directly rather than relying on Title I status alone.

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Contributions

Siddiqa Wassan, lead editor and coordinator, maintained the shared master manuscript and coordinated the editorial workflow across the seven-person team; she integrated revisions, checked consistency across the narrative and tables, and prepared the final manuscript.

Abenezer Girma Beyene, data lead, managed the school-level teacher-return and EOCEP dataset and checked its quantitative consistency, verified the Wade Hampton figures against the school report cards, and reproduced the poverty correlations from the eight school-level poverty values; he also compared English II and Algebra I outcomes across both school years and helped present the findings in the tables.

Aigerim Amandykova reviewed the literature sources and found that the Learning Policy Institute report (Tan et al., 2026) did not support a claim of positive effects from turnover, which led to its use in the literature review as context on turnover patterns.

Mahajabin Chowdhury reviewed and verified evidence and sources, cross-checked supporting data, and examined the Title I classifications and related school-level evidence.

Franciszek Grzyb supported the backup review of project data and quantitative tables.

Anaya Penmetsa reviewed the paper for accuracy, relevance, and consistency, flagged inconsistencies, and reviewed aggregate regional data.

Aidana Seitova reviewed the literature and citations in the literature review, checked the data sources and methodology in the data and methods section, and reviewed the school-level poverty figures and references in Appendix A.

Appendix A. Data Sources

EOCEP results: SCDE School Report Cards, All Students category, English II and Algebra I, 2022–23 and 2023–24. Teacher-return rates: project teacher-return workbook. Poverty: "Percent of students in poverty," South Carolina School Report Cards, Environment → Financial Data, 2023–24. Each row below gives the school's SCDE identifier and its report-card URL.

School	Poverty (%)	SCDE ID	Report-card URL
Berea High School	81.5%	2301002	https://www.screportcards.com/overview/?q=eT0yMDI0JnQ9SCZzaWQ9MjMwMTAwMg
Carolina High School and Academy	86.9%	2301005	https://www.screportcards.com/overview/?q=eT0yMDI0JnQ9SCZzaWQ9MjMwMTAwNQ
Eastside High School	44.3%	2301006	https://www.screportcards.com/overview/?q=eT0yMDI0JnQ9SCZzaWQ9MjMwMTAwNg
Riverside High School	31.4%	2301017	https://www.screportcards.com/overview/?q=eT0yMDI0JnQ9SCZzaWQ9MjMwMTAxNw
Wade Hampton High School	52.0%	2301010	https://www.screportcards.com/overview/?q=eT0yMDI0JnQ9SCZzaWQ9MjMwMTAxMA
J.L. Mann High School Academy of Mathematics, Science, and Technology	43.5%	2301013	https://www.screportcards.com/overview/?q=eT0yMDI0JnQ9SCZzaWQ9MjMwMTAxMw
Mauldin High School	39.1%	2301014	https://www.screportcards.com/overview/?q=eT0yMDI0JnQ9SCZzaWQ9MjMwMTAxNA
Southside High School	78.7%	2301018	https://www.screportcards.com/overview/?q=eT0yMDI0JnQ9SCZzaWQ9MjMwMTAxOA

Source: South Carolina School Report Cards, 2023–24.

Using the eight poverty values above and the corresponding 2023–24 EOCEP C-or-higher rates in Tables I and II, the Pearson correlations are $r = -0.9647$ for English II and $r = -0.9556$ for Algebra I.